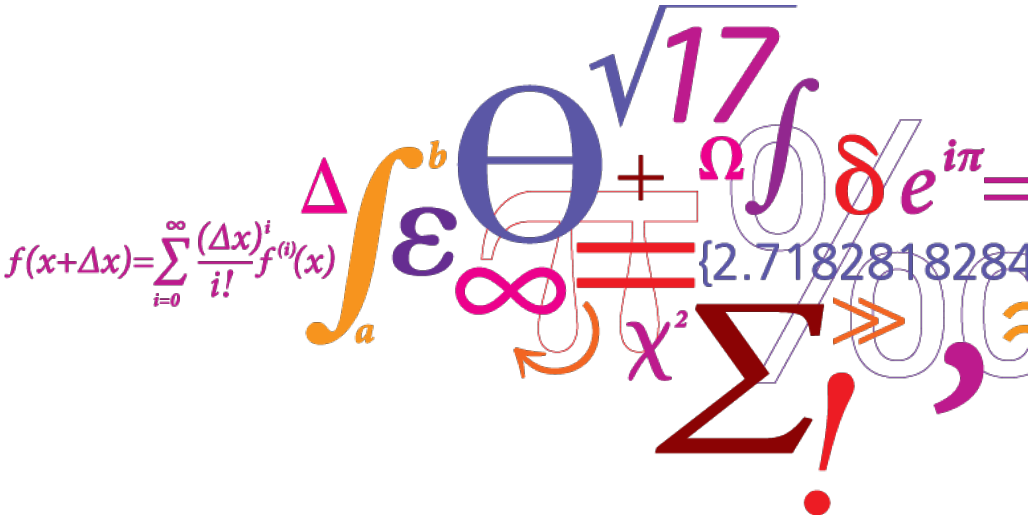
46755 – Renewables in Electricity Markets

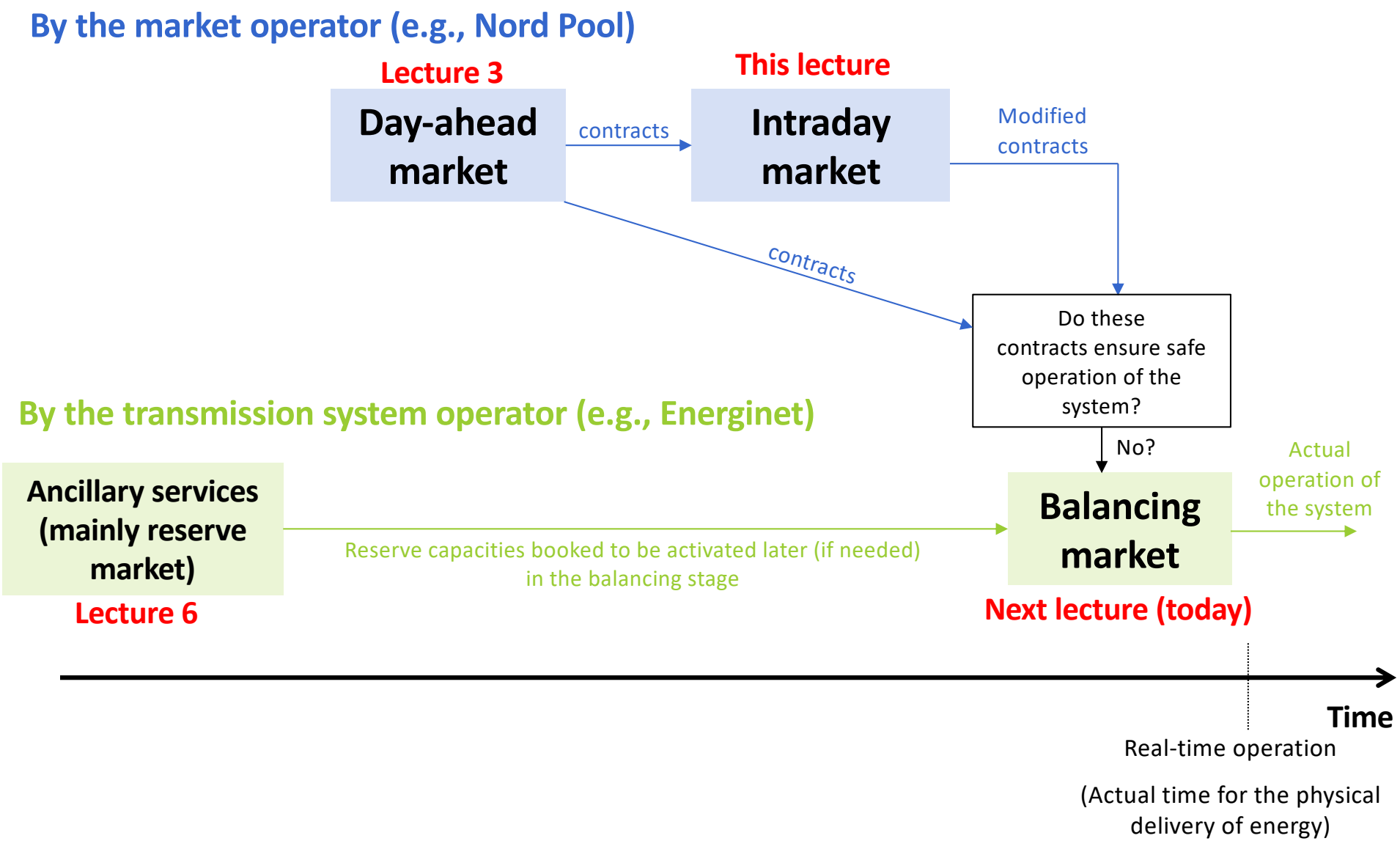
Lecture 5a: Intraday markets

Jalal Kazempour

March 3, 2025



Recap


By the market operator (e.g., Nord Pool)
Lecture 3
Day-ahead market
contracts
This lecture
Intraday market
Modified contracts
contracts
Do these contracts ensure safe operation of the system?
No?
By the transmission system operator (e.g., Energinet)
Ancillary services (mainly reserve market)
Lecture 6
Reserve capacities booked to be activated later (if needed) in the balancing stage
Balancing market
Next lecture (today)
Actual operation of the system
Time
Real-time operation
(Actual time for the physical delivery of energy)

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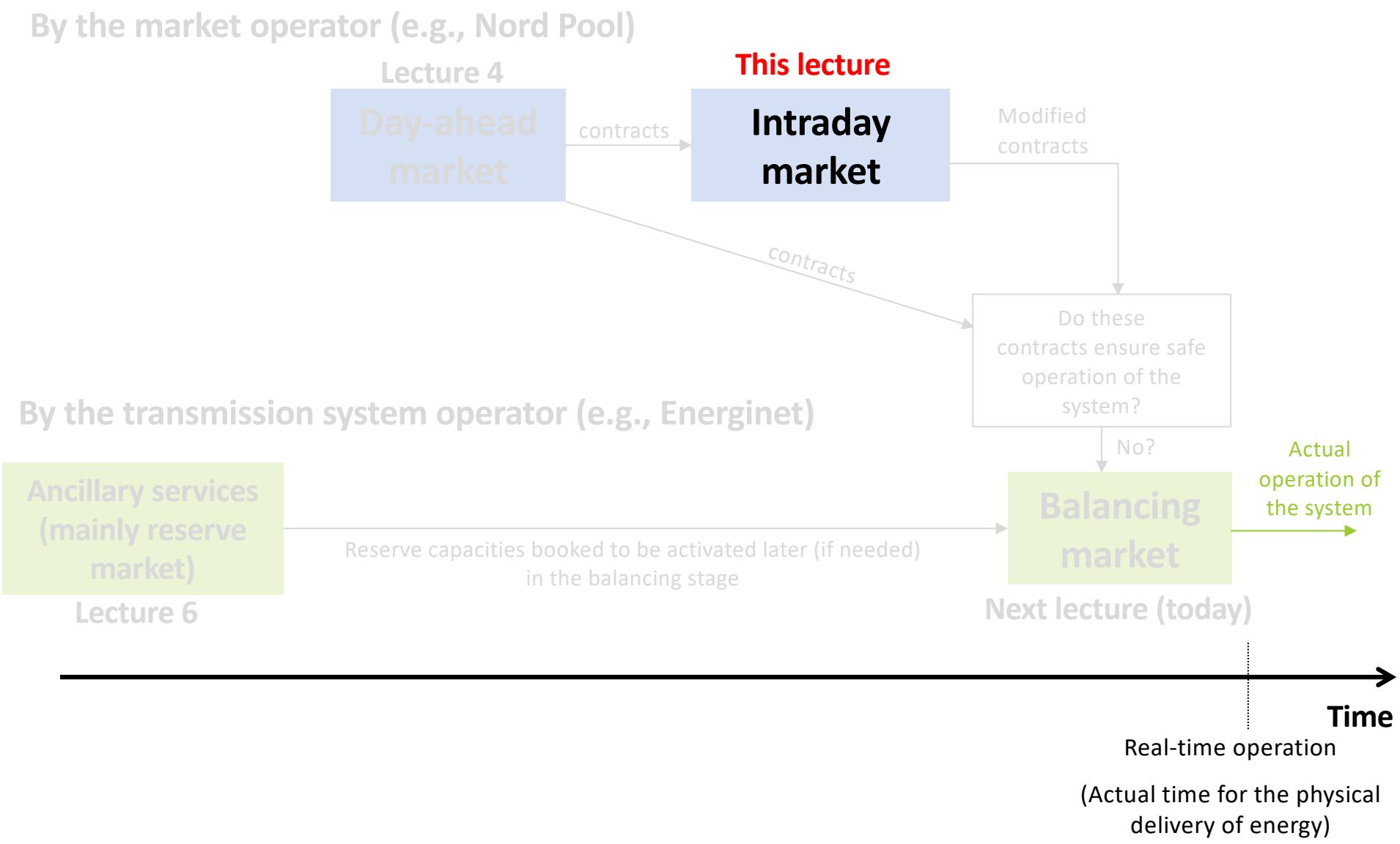
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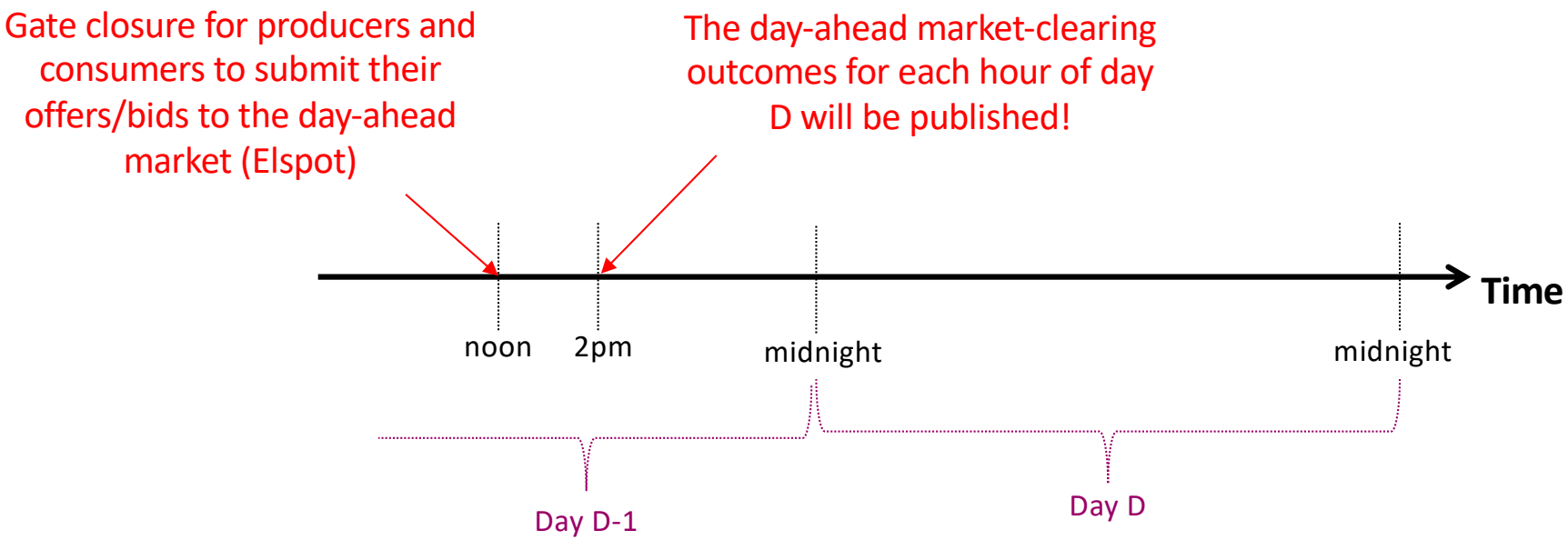
Learning objectives of this lecture

After this lecture, you are expected to:

- Explain the fundamentals of intraday markets in Europe.
- Compare and contrast continuous and auction-based intraday markets.

Timeline

Recall the timeline for the day-ahead market in Europe as discussed in Lecture 3:



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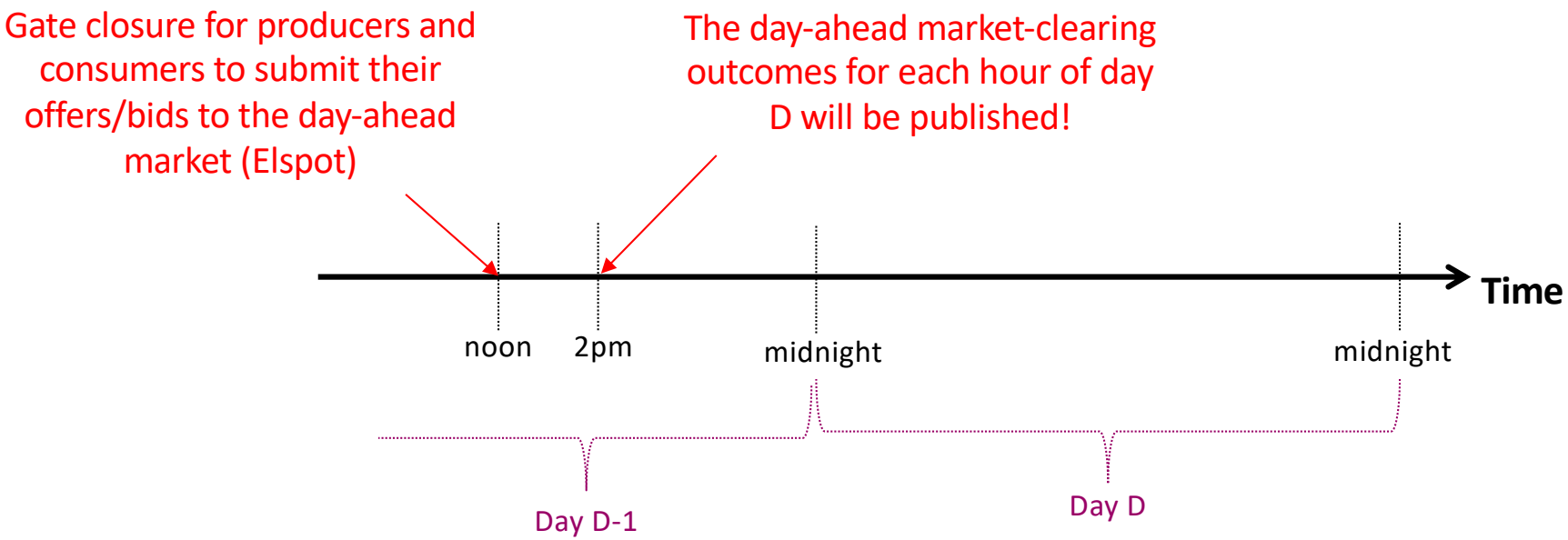

The next market available for market participants is the intraday market!

Why intraday market?

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  - **Updated Forecasts**: They may receive an updated forecast of future renewable power generation or load, which could differ from their earlier forecast.
  - **Unexpected Technical Issues**: Participants may realize that part of their assets will be offline in real-time operation due to technical reasons, even though those assets were already scheduled to produce or consume power in the day-ahead market.
  - **Trading Opportunities**: Market participants may identify new trading opportunities based on changes in market conditions, price signals, or other factors.

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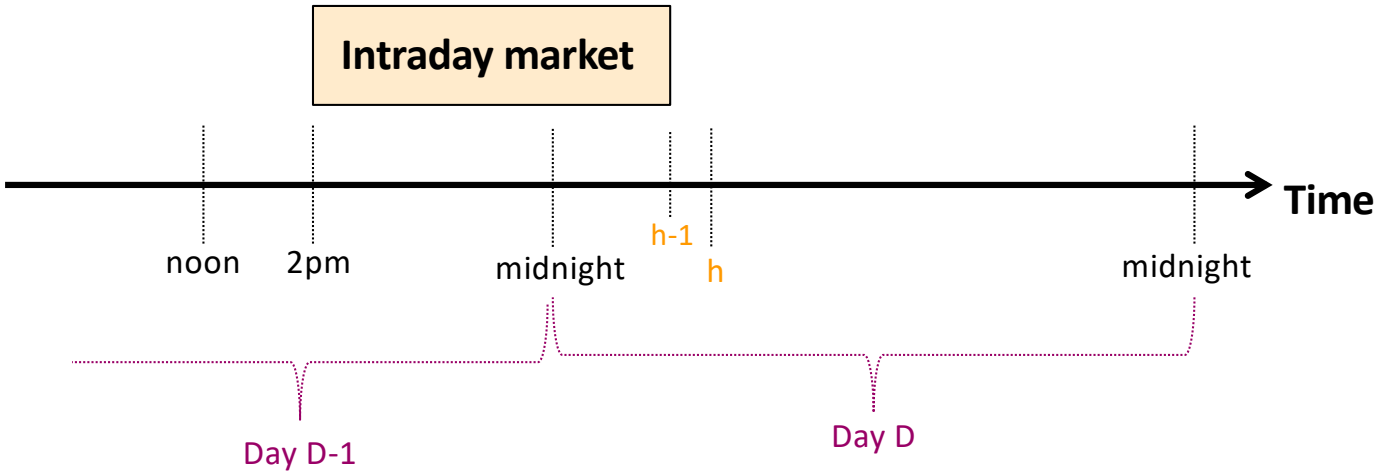
- The intraday market provides the opportunity to offset such deviations, allowing market participants to make adjustments before the balancing market.

Timeline for the intraday market

After publishing the day-ahead market-clearing outcomes at 2 PM on day D-1:

Nord Pool opens the intraday market for each hour h of day D in a **continuous** manner, starting from 2 PM on day D-1 and continuing until hour h-1 of day D.



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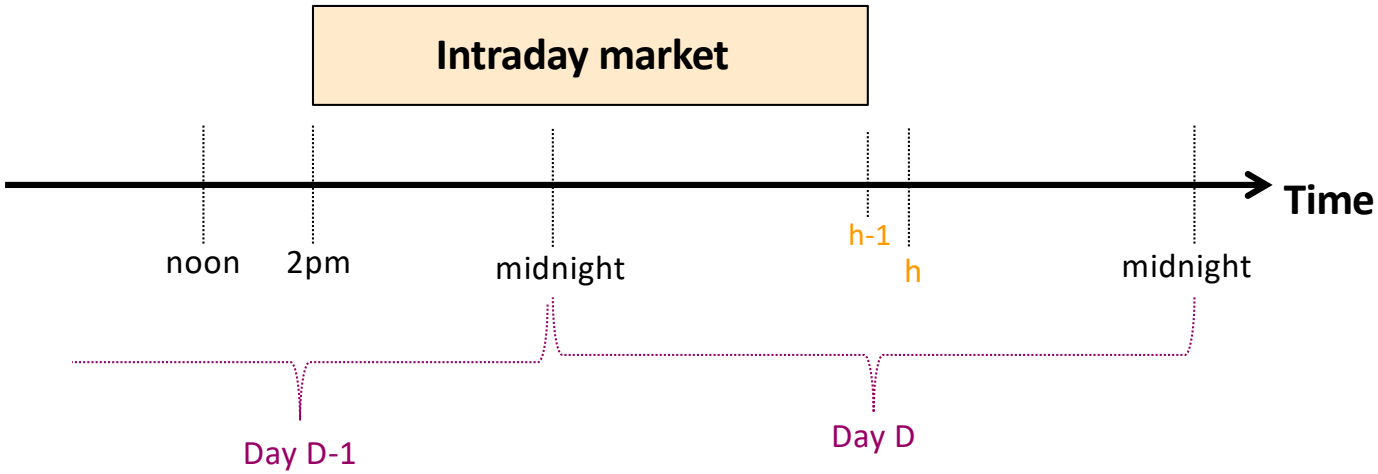

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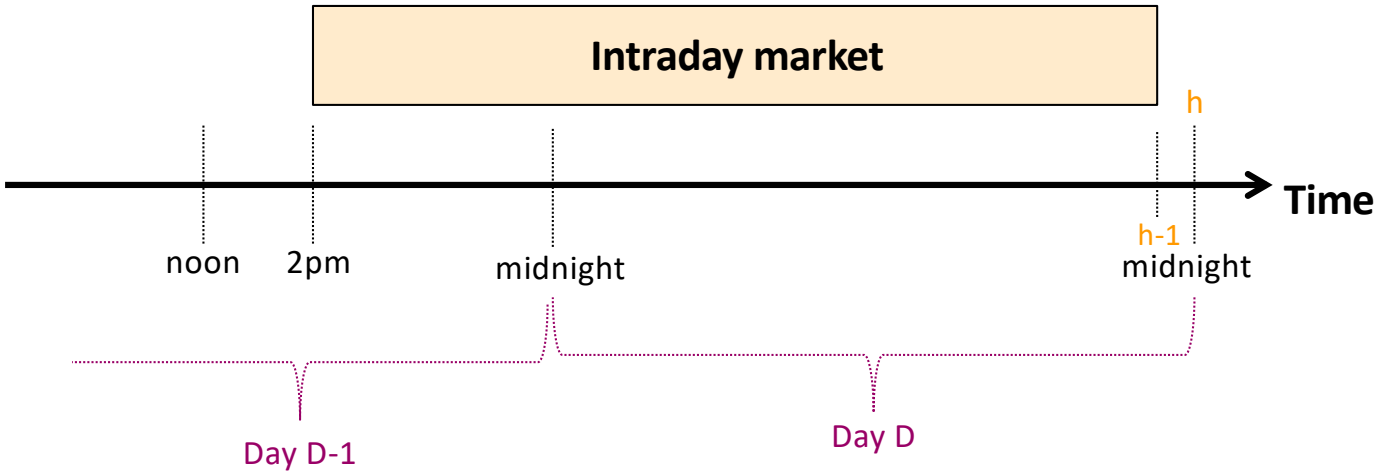

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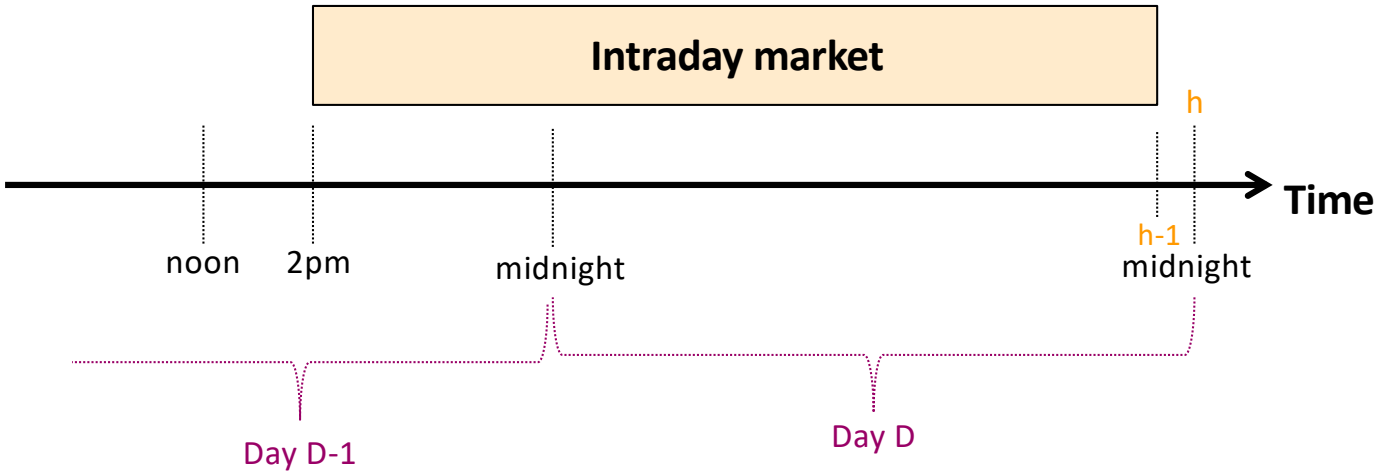

Note:

- The market time unit (granularity) will change soon in 2025 to 15-minute intervals.
- The opening hour may differ in various parts of Europe; for example, it is 3 PM in Germany.
- The market closure (h-1) might also vary across different regions of Europe.

Continuous intraday market

What does a **continuous market** mean?

Created by ChatGPT

Continuous intraday market

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Recall, in the day-ahead market:

- There is a specific gate-closure time (noon) by which power buyers and sellers must submit their bids and offers.
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From noon on day D-1 until hour h-1 on day D, any pair of buyers and sellers can enter into a bilateral contract through the central platform of the intraday market.

This makes the intraday market "continuous"!

Continuous intraday market: Order book

The continuous intraday market is run as an **order book**, with trades cleared as soon as buy and sell orders match

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Order Book			
Buy		**Sell**	
10 MWh	£90 /MWh	10 MWh	£100 /MWh
10 MWh	£80 /MWh	10 MWh	£130 /MWh
		10 MWh	£140 /MWh

All orders are currently unfulfilled as prices do not match

Buy orders with the **highest** price and sell orders with the **lowest** price sit at at the top of the order book

Source: Modo Energy

Continuous intraday market: Order book

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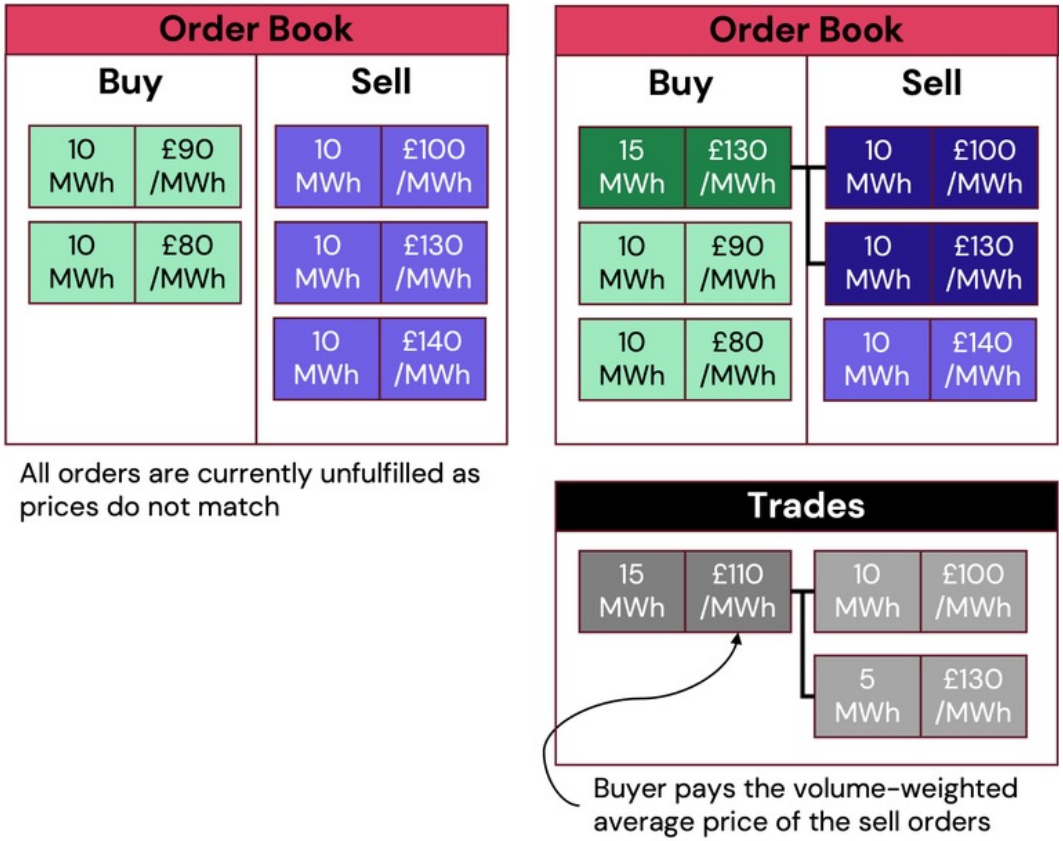

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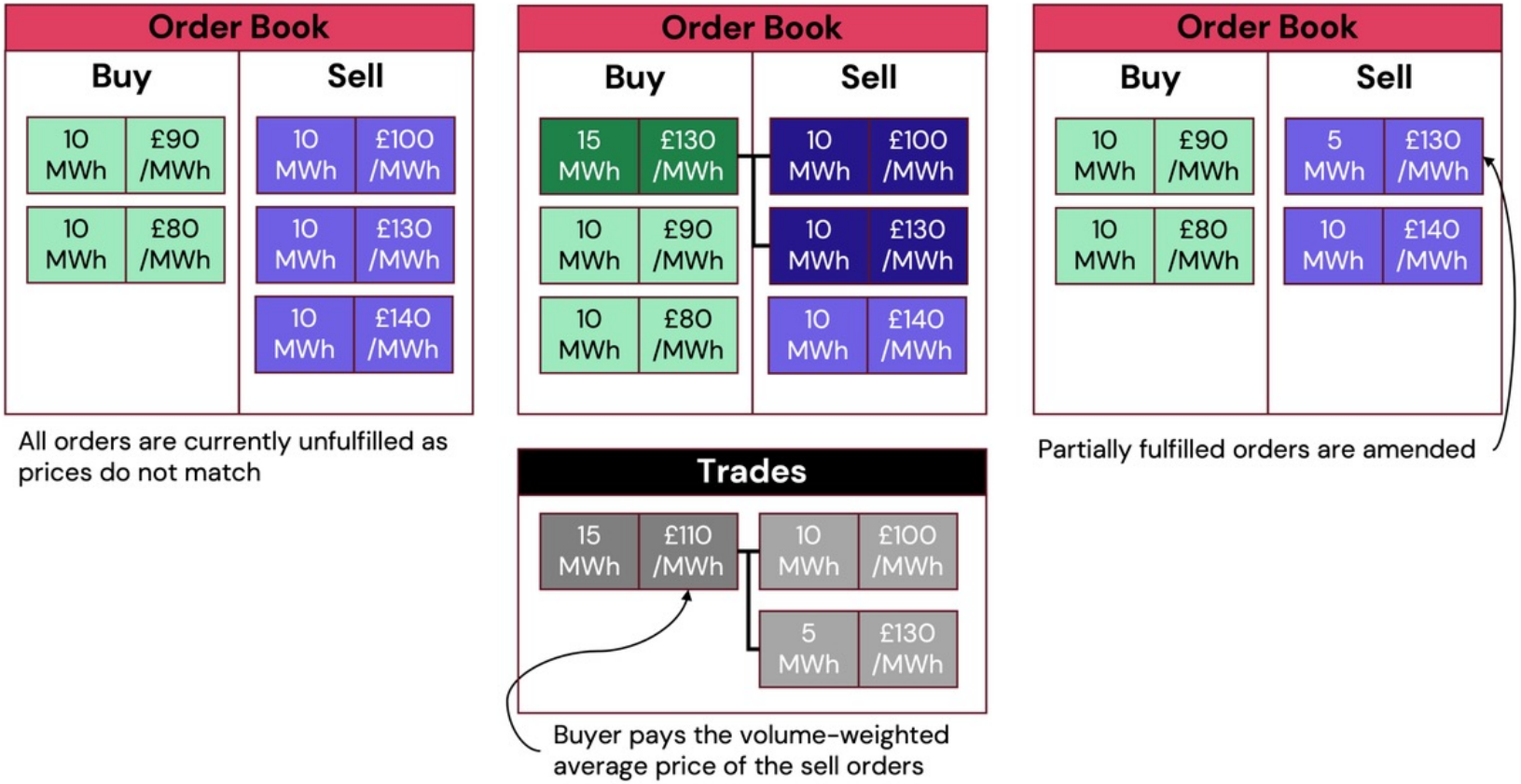


Buy orders with the **highest** price and sell orders with the **lowest** price sit at at the top of the order book

A buy order is received and matches to two sell orders - a **trade is created**

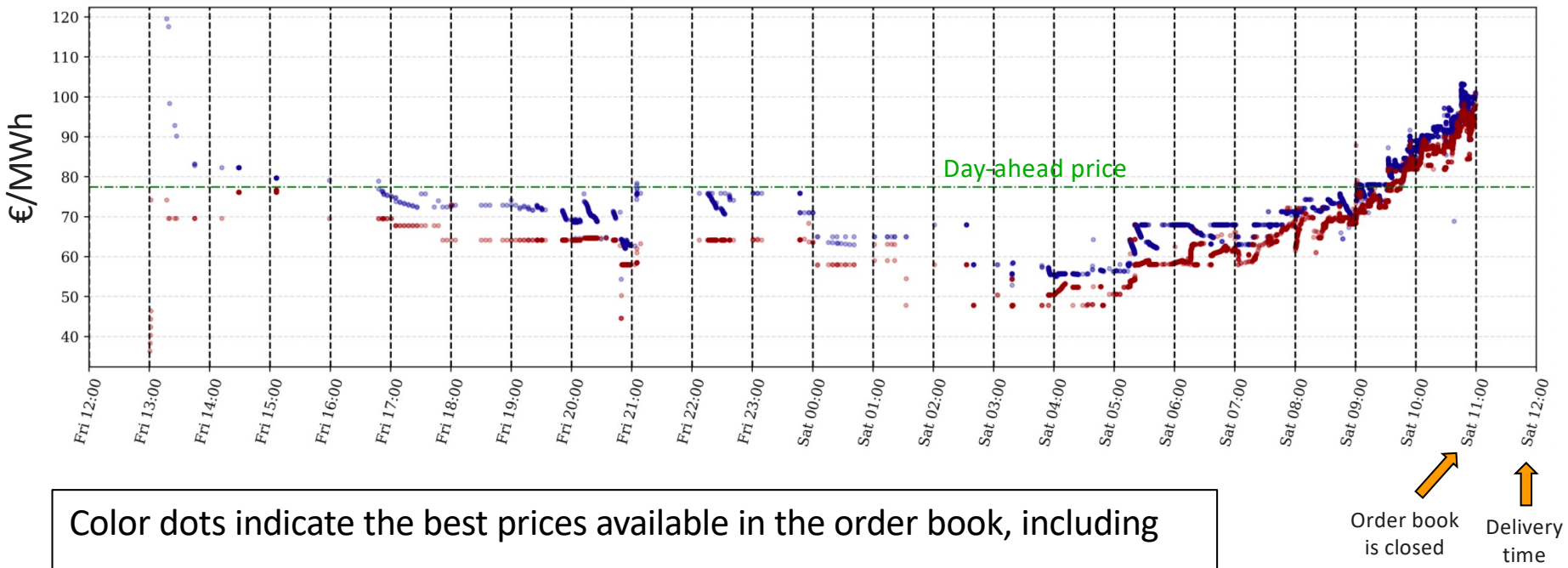
Fulfilled buy and sell orders are removed from the order book – the trade has been cleared

Source: Modo Energy

MODOENERGY

Continuous intraday market: Evolution of trades

Order book for April 15, 2023 (Saturday), 12:00 noon



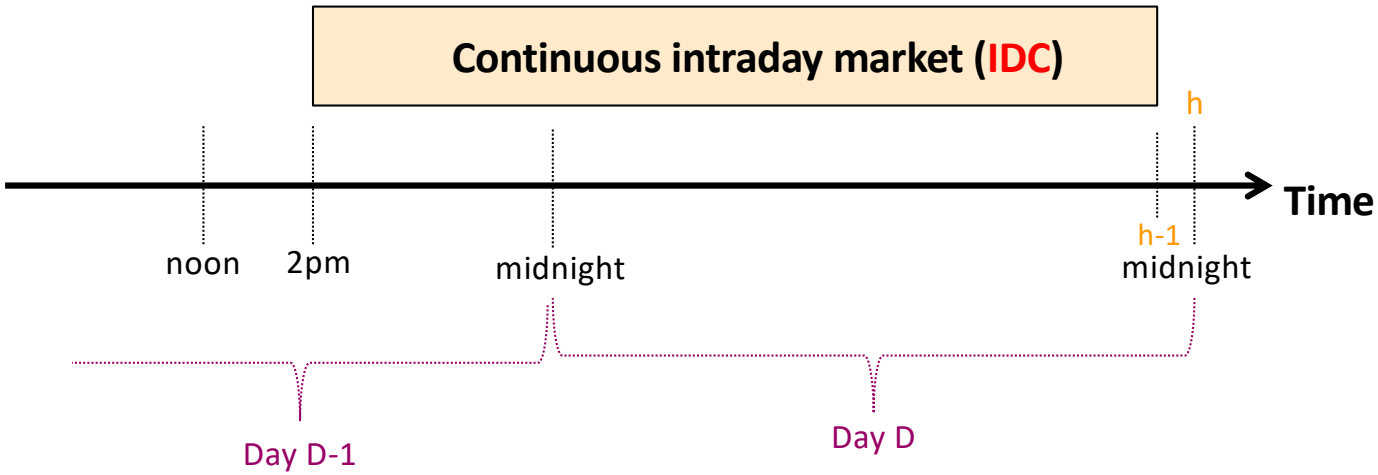
Color dots indicate the best prices available in the order book, including

- The highest **buy price**
- The lowest **sell price**

Note: A dot appears when the previous best price changes.

Credit: Hybrid Greentech and Markus Hvid Monin

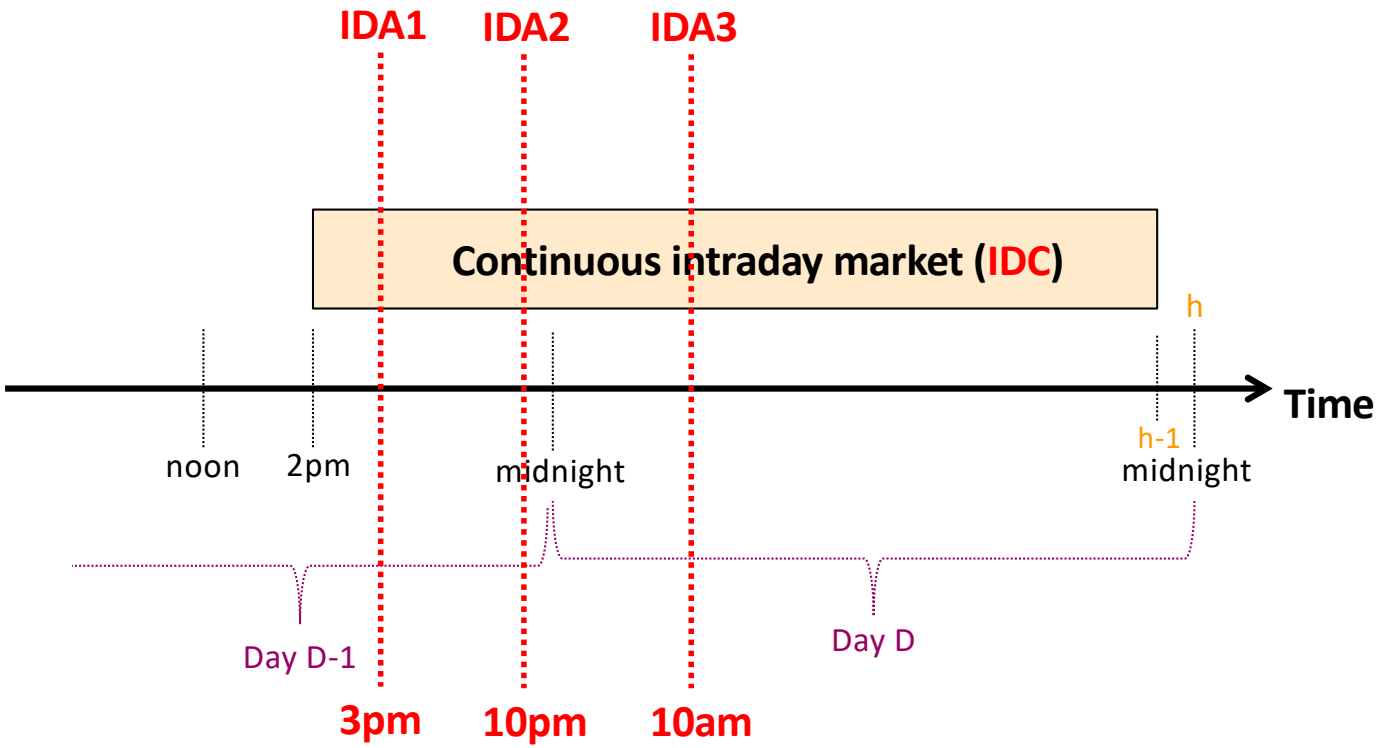
Intraday market: IDC vs. IDA



Intraday market: IDC vs. IDA

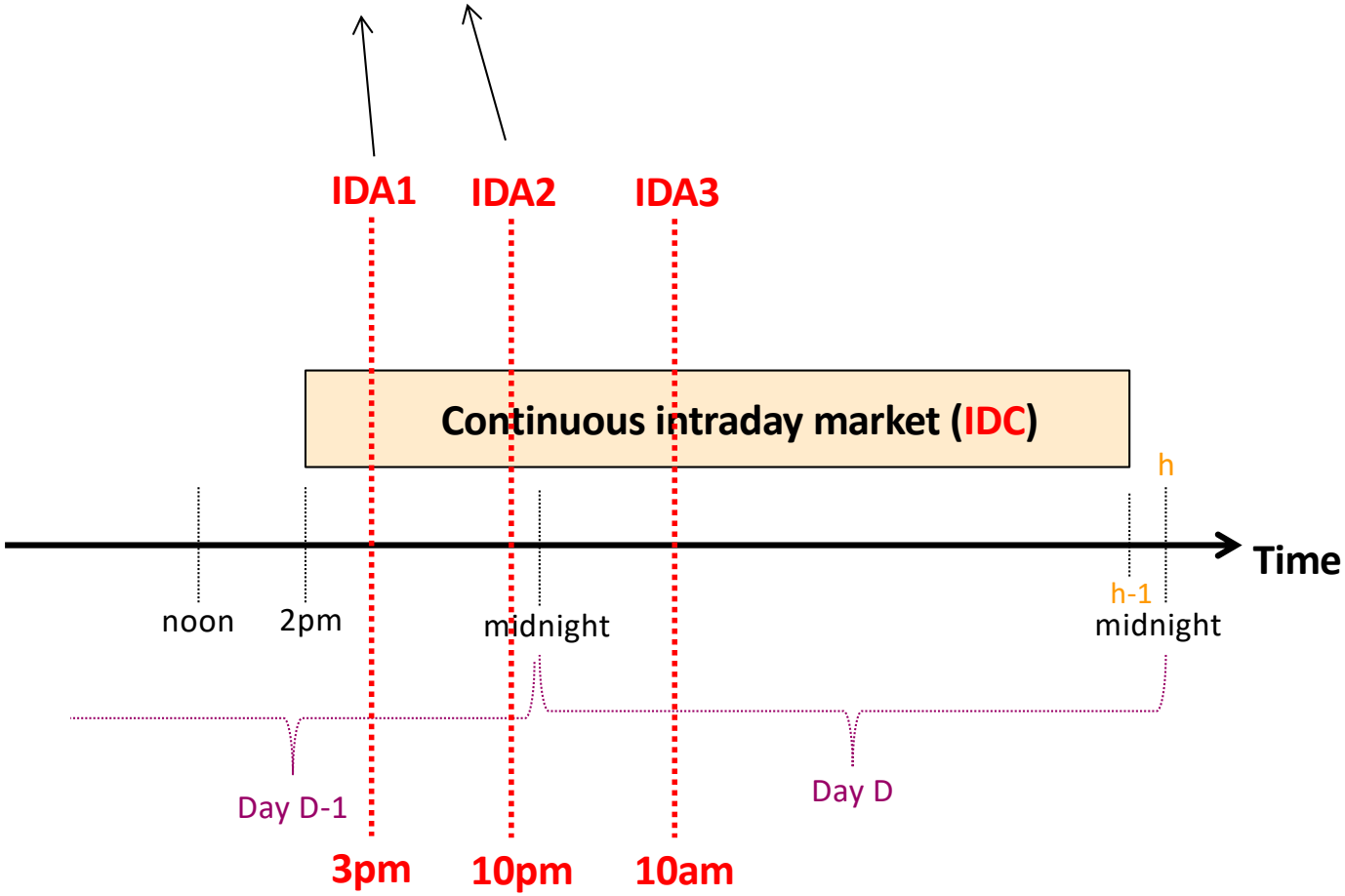
In addition to the **continuous** intraday market (IDC), three **auction-based** intraday markets (IDAs) have recently been introduced in Europe.

- IDA1-IDA3 markets operate with a gate closure and a price formation mechanism based on the intersection of supply and demand curves, similar to the day-ahead market.



Intraday market: IDC vs. IDA

Auctions on day D-1
Trade period: 0:00-24:00 on day D
(hourly granularity, transitioning to 15-minute intervals soon)



Intraday market: IDC vs. IDA

Auctions on day D-1
Trade period: 0:00-24:00 on day D
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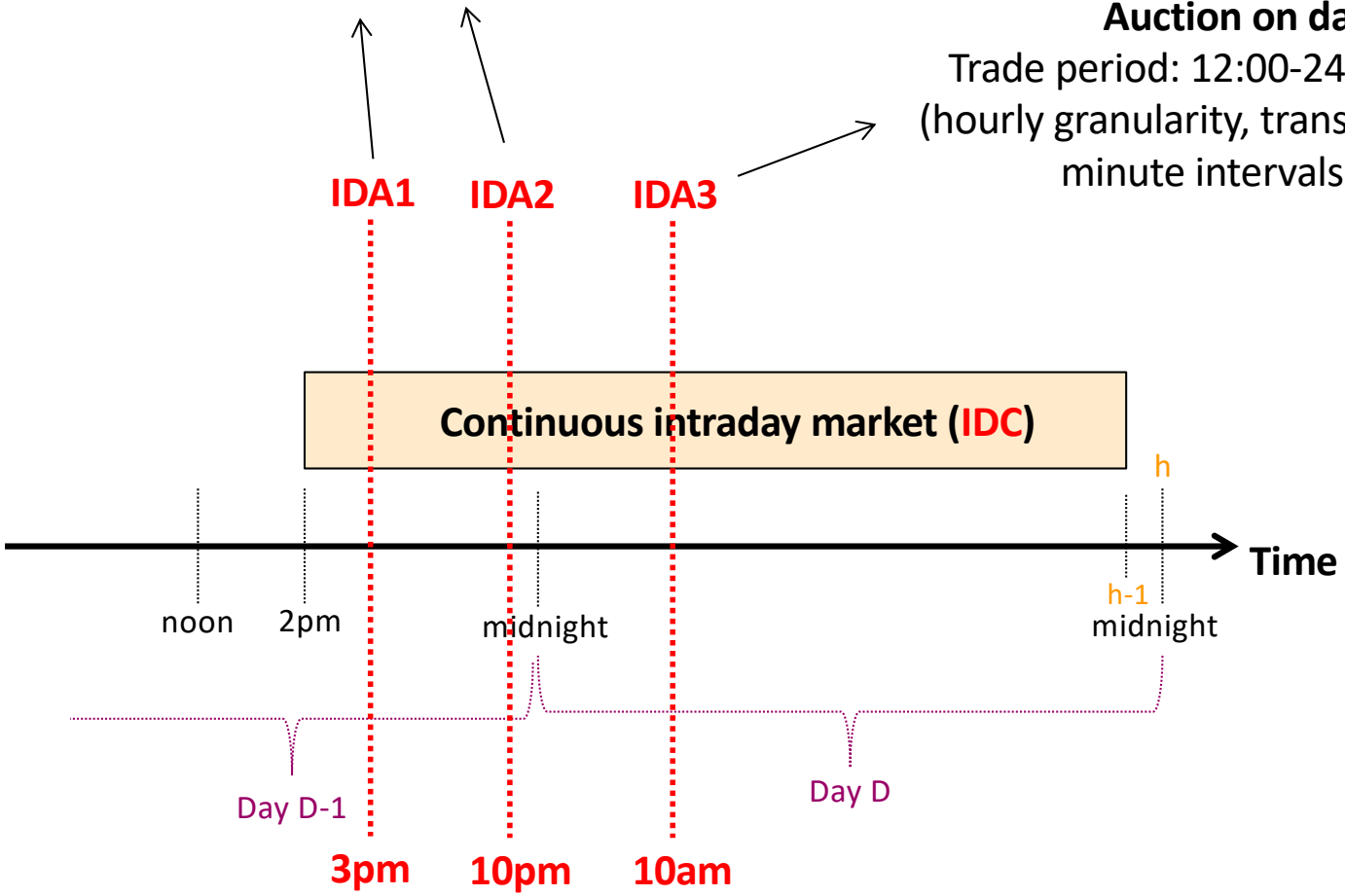
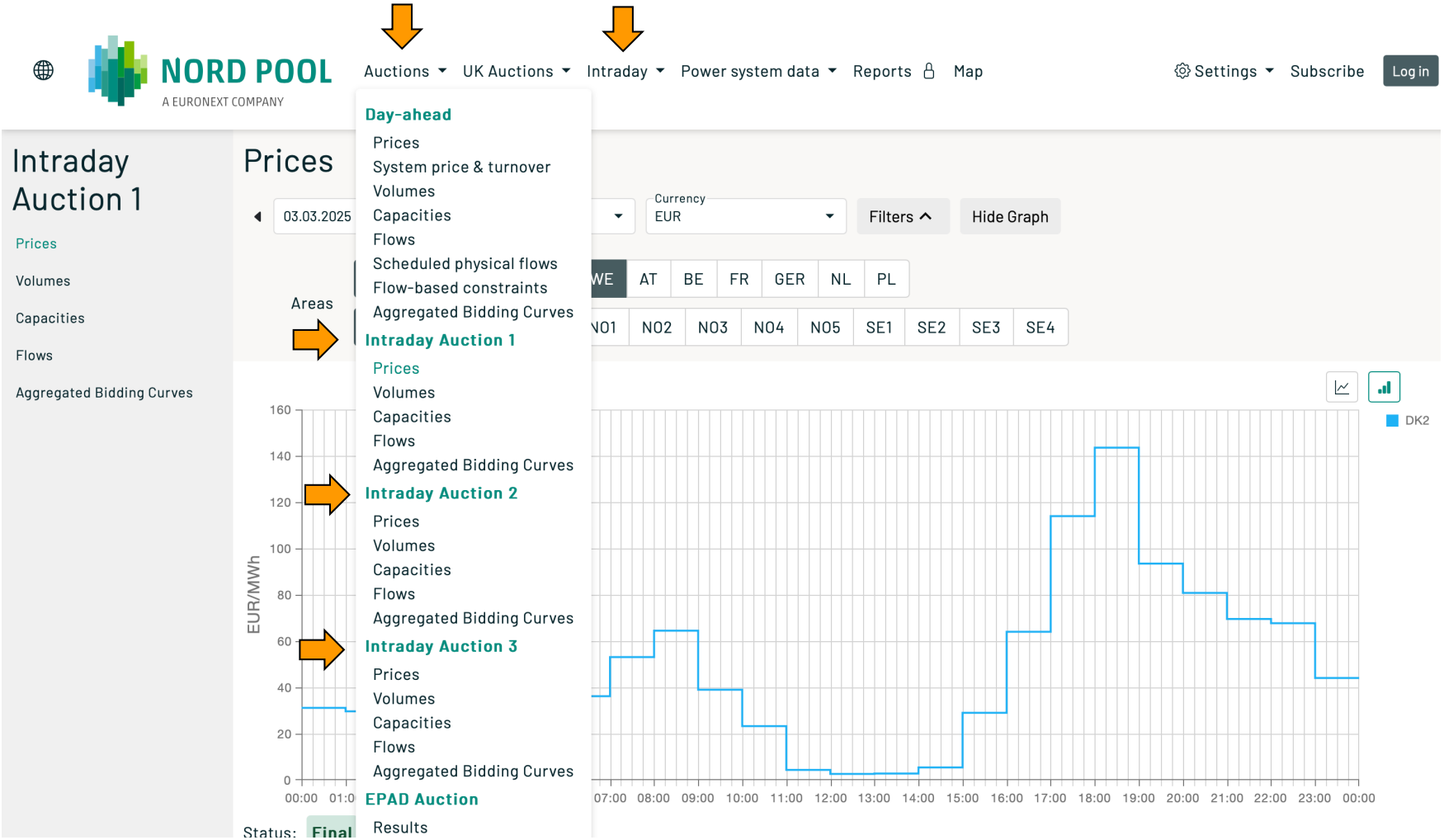
Auction on day D
Trade period: 12:00-24:00 on day D
(hourly granularity, transitioning to 15-minute intervals soon)



Intraday market in Nord Pool

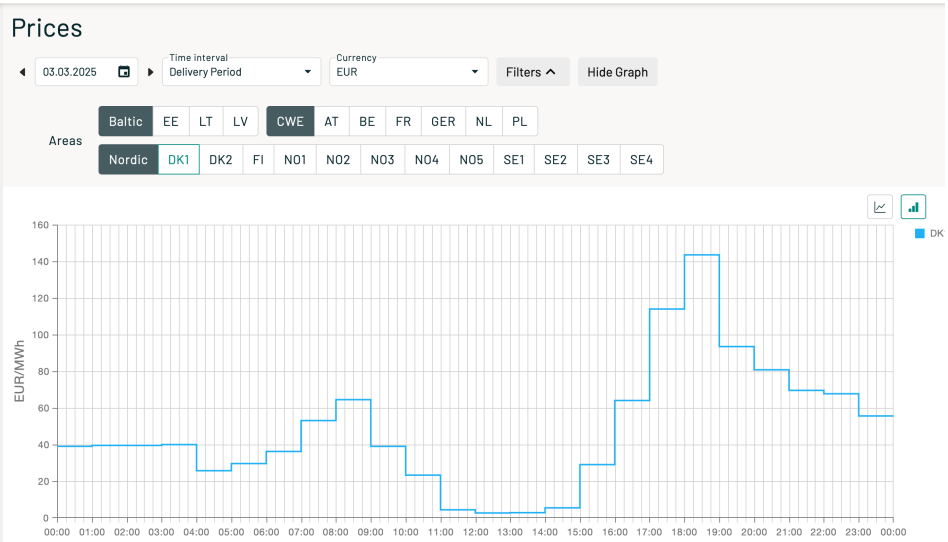
Auctions including **IDA1-IDA3**

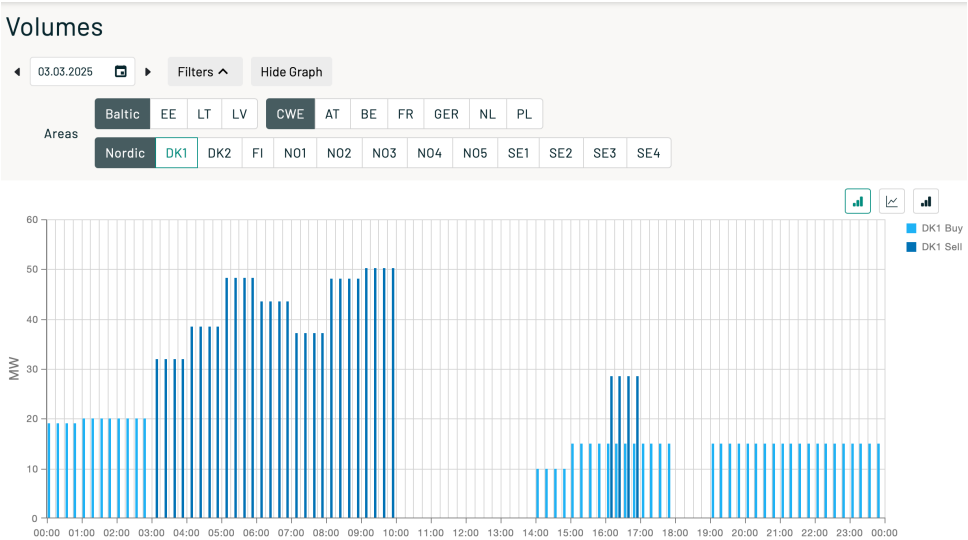
IDC (continuous)

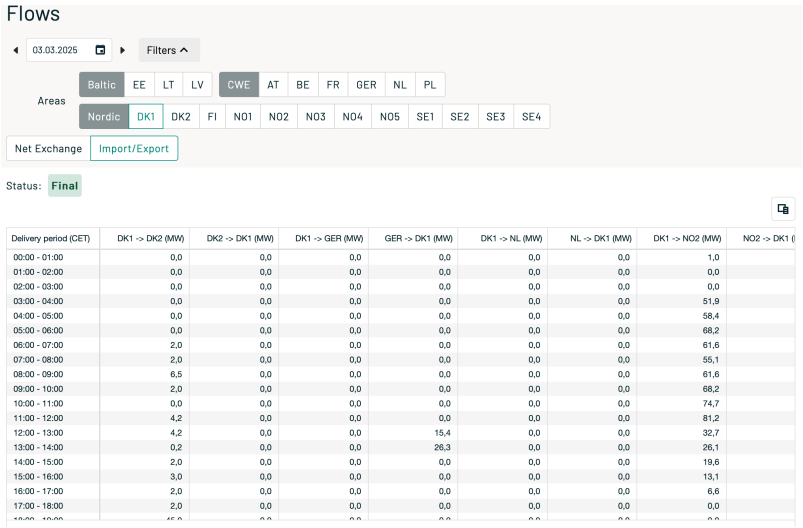


Intraday market in Nord Pool: IDA1

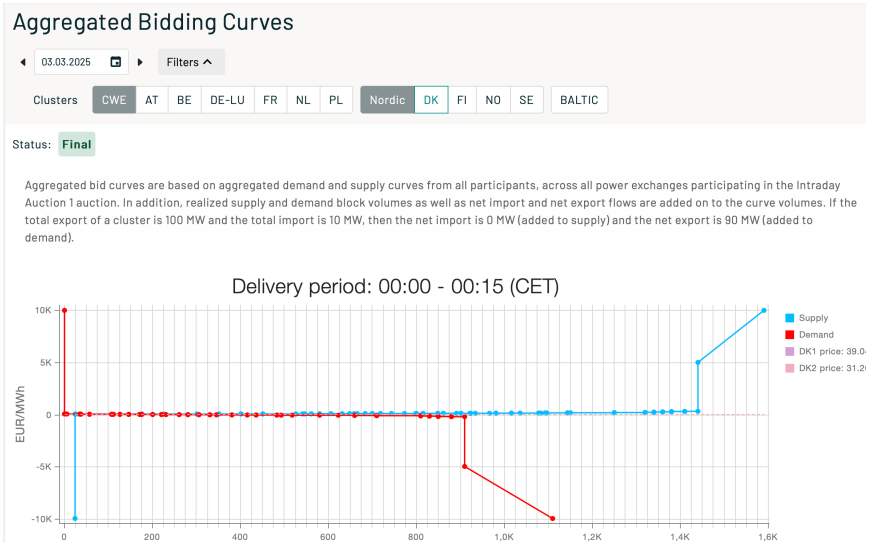
IDA1 market outcomes for DK1 on March 3, 2025 (already with the time granularity of 15 minutes):
Prices, volumes, cross-border flows, and supply-demand curves







Delivery period (CET)	DK1 -> DK2 (MW)	DK2 -> DK1 (MW)	DK1 -> GER (MW)	GER -> DK1 (MW)	DK1 -> NL (MW)	NL -> DK1 (MW)	DK1 -> NO2 (MW)	NO2 -> DK1 (
00:00 - 01:00	0,0	0,0	0,0	0,0	0,0	0,0	1,0	
01:00 - 02:00	0,0	0,0	0,0	0,0	0,0	0,0	0,0	
02:00 - 03:00	0,0	0,0	0,0	0,0	0,0	0,0	0,0	
03:00 - 04:00	0,0	0,0	0,0	0,0	0,0	0,0	51,9	
04:00 - 05:00	0,0	0,0	0,0	0,0	0,0	0,0	58,4	
05:00 - 06:00	0,0	0,0	0,0	0,0	0,0	0,0	68,2	
06:00 - 07:00	2,0	0,0	0,0	0,0	0,0	0,0	61,6	
07:00 - 08:00	2,0	0,0	0,0	0,0	0,0	0,0	55,1	
08:00 - 09:00	6,5	0,0	0,0	0,0	0,0	0,0	61,6	
09:00 - 10:00	2,0	0,0	0,0	0,0	0,0	0,0	68,2	
10:00 - 11:00	0,0	0,0	0,0	0,0	0,0	0,0	74,7	
11:00 - 12:00	4,2	0,0	0,0	0,0	0,0	0,0	81,2	
12:00 - 13:00	4,2	0,0	0,0	15,4	0,0	0,0	32,7	
13:00 - 14:00	0,2	0,0	0,0	26,3	0,0	0,0	26,1	
14:00 - 15:00	2,0	0,0	0,0	0,0	0,0	0,0	19,6	
15:00 - 16:00	3,0	0,0	0,0	0,0	0,0	0,0	13,1	
16:00 - 17:00	2,0	0,0	0,0	0,0	0,0	0,0	6,6	
17:00 - 18:00	2,0	0,0	0,0	0,0	0,0	0,0	0,0	



Intraday market in Nord Pool: IDC

IDC market outcomes for DK1 on March 1, 2025

- **High Price** = max of prices
- **Low Price** = min of prices
- **VWAP** (Volume Weighted Average Price) = (sum of price*volume_in_MWh)/(sum of volumes_in_MWh)
- **VWAP1H** (Volume Weighted Average Price for trades done after 1h before *delivery start time*) = (sum of price*volume_in_WWh)/(sum of volumes_in_WWh) where trades meet the condition: trade time + 1 hour ≥ delivery start time
- **VWAP3H** (Volume Weighted Average Price for trades done after 3h before *delivery start time*) = (sum of price*volume_in_MWh)/(sum of volumes_in_MWh) where trades meet the condition: trade time + 3 hours ≥ delivery start time

Guest talk at DTU by David Wozabal

David Wozabal: Algorithmic Trading in Continuous Intraday Power Markets

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Speaker: David Wozabal (Vrije Universiteit Amsterdam)
Invited talk at the Technical University of Denmark (DTU), Copenhagen, June 28, 2024 ...more

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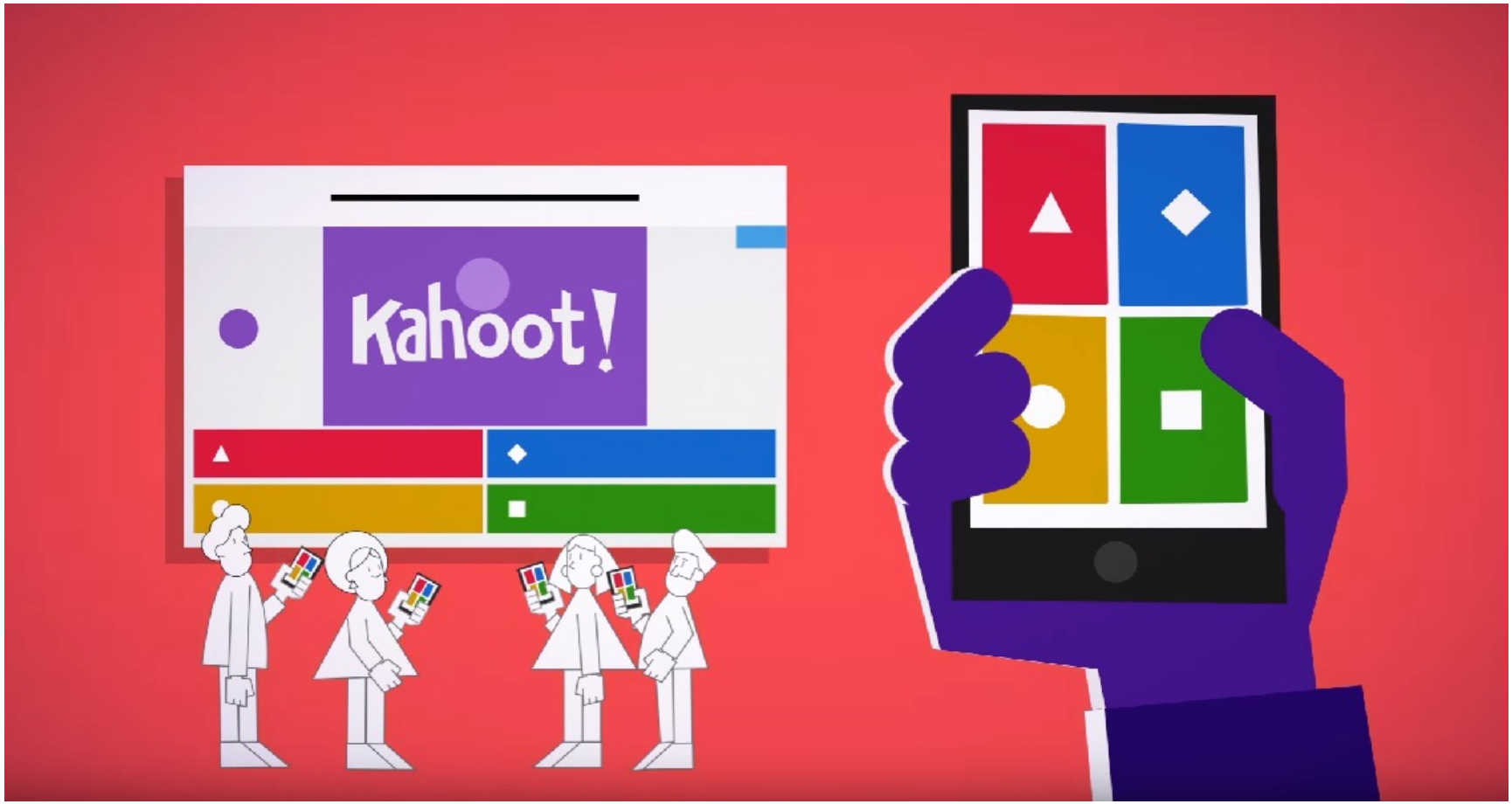
Kahoot!

Thanks for your attention!

Email: jalal@dtu.dk